

Alston Daniel Mendonca

AI/ML Engineer | Generative AI | Computer Vision | Backend Systems

+971 56 3243787 | alstondmendonca@gmail.com | linkedin.com/in/alstonmendonca

github.com/alstonmendonca | alston.vercel.app

Dubai, UAE | Open to UAE roles

SUMMARY

AI/ML Engineer with hands-on experience building and deploying generative AI, NLP, and computer vision systems in production-like environments. Strong in Python, FastAPI, PyTorch, TensorFlow, Hugging Face, AWS Bedrock, and end-to-end model-serving workflows. Built production-oriented ML pipelines and AI-enabled applications with a focus on scalability, reliability, and business outcomes.

EXPERIENCE

Co-Founder and CTO

Apr 2026 – Present

ViperCore

Mangaluru, India

- Co-founded and shipped an offline-first restaurant POS platform adopted by 50+ cafes across Karnataka.
- Architected backend and data workflows for billing, inventory, reporting, and sync-safe local-first operation in low-connectivity environments.
- Drove product roadmap, release cycles, and customer feedback loops to improve stability, feature adoption, and operational reliability.

Backend Developer and Machine Learning Engineer Intern

Nov 2025 – Mar 2026

Aquera

Bengaluru, India

- Built and improved ML pipelines for Identity and Access Management (IAM), focused on service-account discovery and classification.
- Fine-tuned BERT-based text classifiers and evaluated model behavior using precision-recall tradeoffs for imbalanced, weakly labeled data.
- Applied CatBoost, LightGBM, RuleFit, LLM-assisted features, and Elkan-Noto positive-unlabeled learning for production-oriented classification tasks and model evaluation.

Software Developer and AI Intern

Oct 2024 – Oct 2025

Heritage Safety Training Centre

Dubai, UAE

- Built and launched the corporate website using React and Vite to improve accessibility, navigation, and content maintainability.
- Developed a full-stack Learning Management System for training operations, course workflows, and administrative tracking.
- Implemented an LLM-powered support chatbot to automate responses for recurring learner and training-related queries.

SELECTED PROJECTS

DressGuard AI | *Python, FastAPI, YOLOv8, CUDA, Redis, React, Vite*

- Led a team of 4 to deliver a real-time PPE/clothing compliance system across 4 input modes: webcam, IP camera, image, and video.
- Fine-tuned YOLOv8 models on custom datasets for domain-specific detection and confidence-calibrated alerts.
- Built a configurable rule engine, model-selection hooks, and visual overlays for production-style real-time inference pipelines.
- Integrated face detection and recognition with FaceNet and Redis for low-latency identity lookups.

ProperLCP | *Electron, React, Node.js, SQLite, Git*

- Led a team of 2 to design and deploy a POS system used in live campus restaurant operations.
- Delivered 75+ features spanning billing, inventory tracking, order management, and reporting dashboards.
- Operated in production at The Lassi Corner, SJEC, with direct user feedback and iterative improvements.

NurseMindfulnessStudyApp | *React Native, TypeScript, Supabase*

- Built a mobile app for 3 structured modules: mental health survey capture, emotional check-ins, and journal workflows for a research use case.
- Implemented data collection flows and backend integration to support reliable participant submissions and tracking.

EDUCATION

St. Joseph Engineering College <i>Bachelor of Engineering in Artificial Intelligence and Machine Learning CGPA: 9.58</i>	Mangaluru, India <i>Aug 2022 – Apr 2026</i>
GEMS Education <i>High School (Computer Science) Grade 12 Board Exams: 95%</i>	Dubai, UAE <i>Apr 2013 – Mar 2022</i>

TECHNICAL SKILLS

ML and AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, BERT, YOLOv8, LLM Applications, CatBoost, LightGBM, Pandas, NumPy
GenAI and NLP: Prompt Engineering, LLM Application Development, Model Evaluation
Programming: Python, TypeScript, JavaScript, SQL, Java, C/C++, R
Backend and MLOps: FastAPI, Node.js, Express.js, REST APIs, Docker, Redis, SQLite, AWS Bedrock, Boto3, CUDA, CI/CD
Frontend and Tools: React, Next.js, Vite, Electron, Git, Supabase